

Abdelrahman ElShafay

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EDUCATION

Carleton University

B.C.S Honours Computer Science, specializing in AI and ML (CO-OP)

Ottawa, ON

Sept 2022 – May 2028

- 2024 – 2025 Dean's List Award
- CGPA: 10.08/12.00, Major CGPA: 10.50/12.00

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQLite, JavaScript, TypeScript, HTML/CSS, R, TCL

Technologies: Angular, Node.js, Rest APIs, I^2C , Linux Kernel, Perfetto, Raspberry Pi

Developer Tools: Git, Docker, VS Code, IntelliJ, Qt Creator, BitBucket, Jira, Jenkins

WORK EXPERIENCE

Ford Motors Company

Embedded Software Engineer Coop | C++, Python, Git, Jira

Kanata, ON

Incoming Fall 2026

- Will contribute to the design, implementation, and testing of embedded software for telematics and in-vehicle communication systems.

Solace

Systems Software Engineer Coop | C/C++, Linux, TCL, Git, Bash, Perfetto

Kanata, ON

Jan 2026 – April 2026

- Implemented and reviewed C++ changes to the PubSub+ message-spool persistence layer on Linux, validating durability, replay, and recovery behaviour through restart and failover testing using TCL.
- Profiled spool throughput and latency under load using Perfetto performance testing, and collecting traces/logs, then shipped C++ concurrency and locking improvements on Linux using pthreads and mutexes to reduce contention by **25%** and keep guaranteed-delivery workflows low-latency at scale.
- Placed **2nd** at Hack V - Into the Deep, Solace's company-wide hackathon. Built a cross-session context feature on SAM (Solace Agent Mesh) using React, Python FastAPI, and BM25 search to eliminate repetitive context re-entry that costs developers **~10-20%** of their working time.

Ribbon Communications

Frontend Software Engineer Coop | Angular, REST, Node.js, jQuery, Bootstrap

Kanata, ON

Sept 2025 – Dec 2025

- Engineered a reusable Angular components for form validation, error handling, and user alerts, standardizing form behavior across multiple application workflows and reducing duplicate frontend implementations.
- Resolved multiple UI/UX bugs and enhancement tickets in Jira, improving user interface and frontend-backend integration stability and contributing to a **faster** feature delivery rate within agile sprint cycles.
- Developed Angular services and API integration logic for an AI chatbot interface, managing asynchronous requests, response handling, and frontend state updates between the UI and backend endpoints.

Carleton University

Undergraduate Teaching Assistant | C++, Linux, Data structures & Algorithms

Ottawa, ON

Jan 2025 – April 2025

- Leading weekly tutorial sessions for **600+ students** designed to reinforce course material, provide in-depth explanations of C++ concepts, and guide students through hands-on problem-solving activities to enhance their understanding and application of programming principles.
- Designed and implemented a grading automation tool that reduced evaluation time by **64%**, streamlining workflow and improving turnaround for student feedback.

PROJECTS

BMP280 Linux Kernel Driver | C, Linux Kernel, I^2C , Raspberry Pi

June 2025 – July 2025

- Developed a Linux kernel I^2C device driver for the BMP280 sensor on a Raspberry Pi, exposing temperature and pressure readings through sysfs.
- Configured BMP280 control, status, and calibration registers to read raw sensor data and convert measurements using datasheet compensation formulas.
- Implemented kernel-space sensor initialization, register reads, and error handling to support reliable communication over the I^2C bus.